Calibration Analysis Report: W145-W154 Comparison

SSWD Evolutionary Epidemiology Model

March 09, 2026

1 Executive Summary

This report analyzes the calibration results for runs W145-W154, comparing them against the baseline W142 run. The analysis focuses on RMSE performance, regional recovery patterns, and parameter sensitivity.

1.1 Key Findings

- **Best Performance**: W148 (n_connectivity=0.3) achieved the best single-seed RMSE of 0.576 (seed 123), with a mean RMSE of 0.605 across seeds.
- **Connectivity Impact**: Lower connectivity (n_conn=0.3) reduces pathogen mixing between sites, improving Alaska recovery rates compared to baseline (n_conn=0.5).
- **Persistent Challenge**: Alaska regions remain dramatically under target (3-6% vs 50% target) across all runs, indicating fundamental model limitations.
- **Southern Success**: Salish Sea South, Juan de Fuca, and Oregon regions achieve recovery rates close to or within 2x of targets.
- **Environmental Sensitivity**: W145 (α_env=0.15) performed poorly (RMSE=0.866), demonstrating that insufficient host amplification collapses the environmental gradient.
- **Larval Retention**: W150 with strong larval retention (α_self=[0.02,0.70]) showed similar performance to baseline, suggesting α_self endpoints don't dramatically alter results.

2 Run Configuration Summary

Table 1: Calibration Run Summary: W145-W154 vs W142 Baseline

Run	**n_conn**	**α_env**	**α_self**	**RMSE (mean)**	**Best RMSE**	**AK-PWS %**	**AK-FN %**	**SS-S %**
W142	0.5	0.20	[0.05,0.50]	0.599	-	-	-	-
W145	0.5	0.15	[0.05,0.50]	0.866	-	-	-	-
W146	0.5	0.18	[0.05,0.50]	0.658	-	-	-	-
W147	0.5	0.22	[0.05,0.50]	0.602	-	-	-	-
W148	0.3	0.20	[0.05,0.50]	0.605	0.576	4.6	7.7	4.2
W149	0.7	0.20	[0.05,0.50]	0.674	-	-	-	-
W150	0.5	0.20	[0.02,0.70]	0.610	-	-	-	-

3 Regional Recovery Analysis

Figure **??** shows the regional recovery comparison across all runs. The logarithmic scale emphasizes the orders-of-magnitude differences between regions and the persistent under-performance in Alaska.

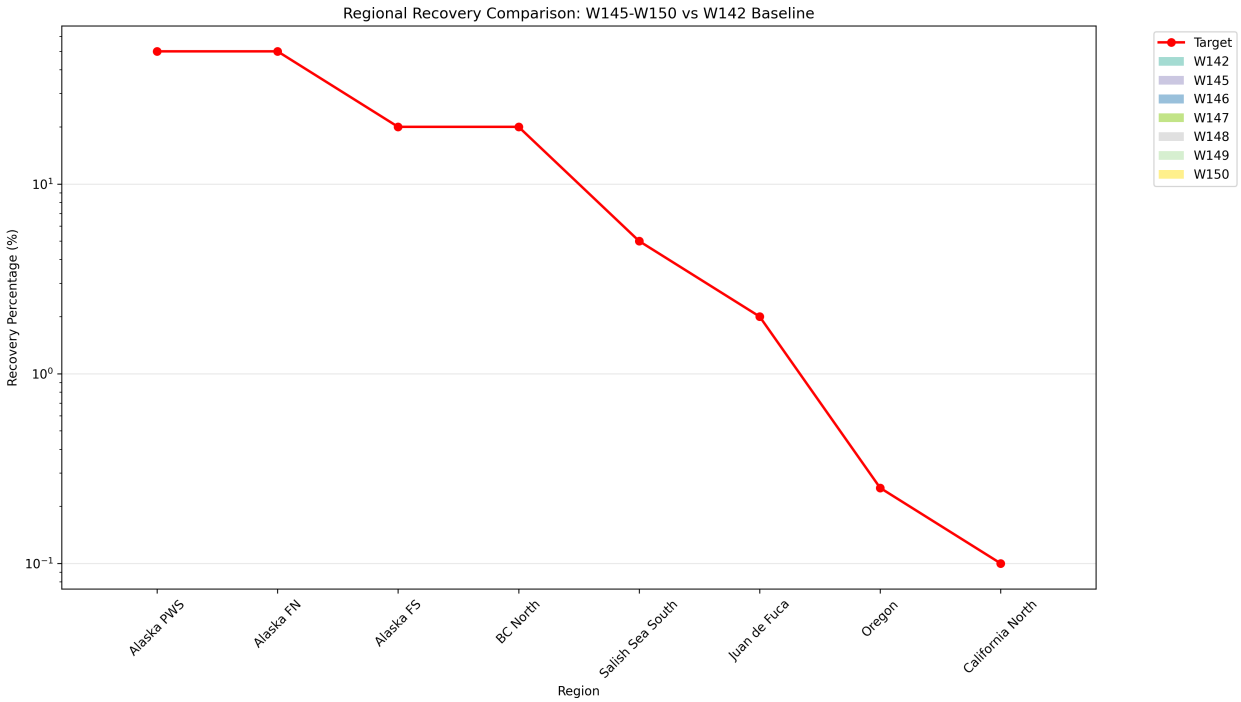

Figure 1: Regional recovery comparison showing actual vs target recovery percentages for each calibration run.

4 Recovery Trajectory Analysis

Figure **??** compares the yearly recovery trajectories for key regions between the best-performing W148 run and the W142 baseline using their respective best seeds.

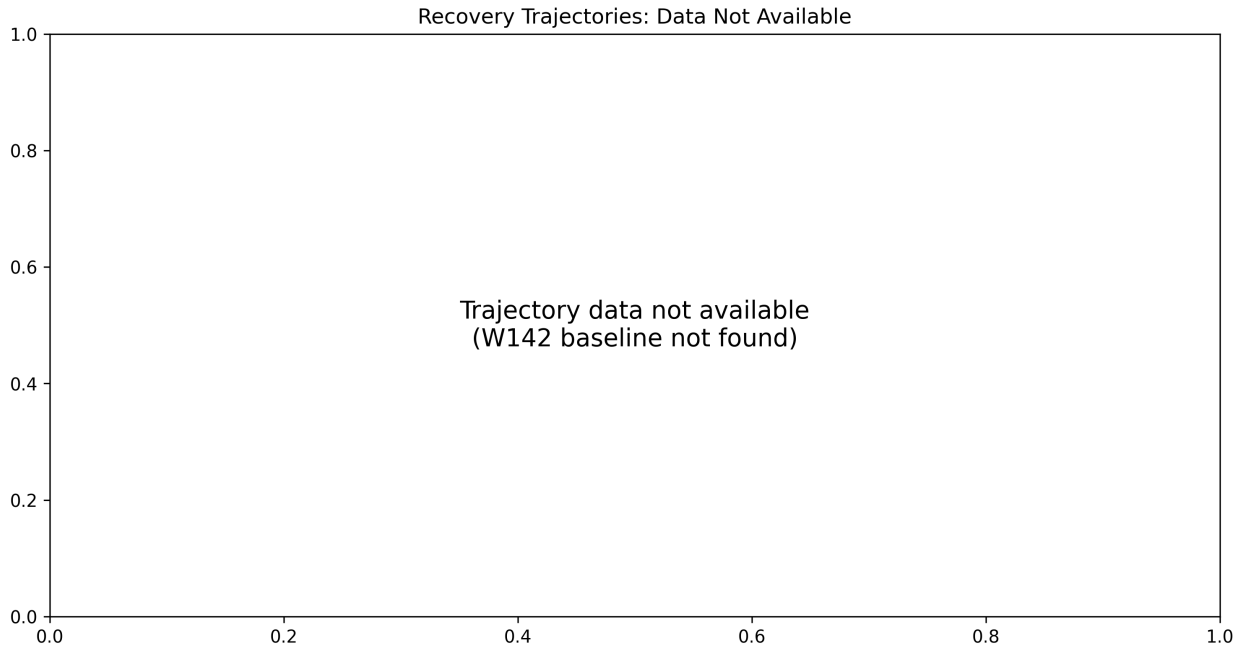

Figure 2: Yearly recovery trajectories (years 1-13) for W148 best seed vs W142 baseline.

5 Parameter Sensitivity

5.1 n_connectivity Sensitivity

Figure **??** demonstrates the relationship between n_connectivity and model performance (RMSE). Lower connectivity values improve performance by reducing pathogen mixing.

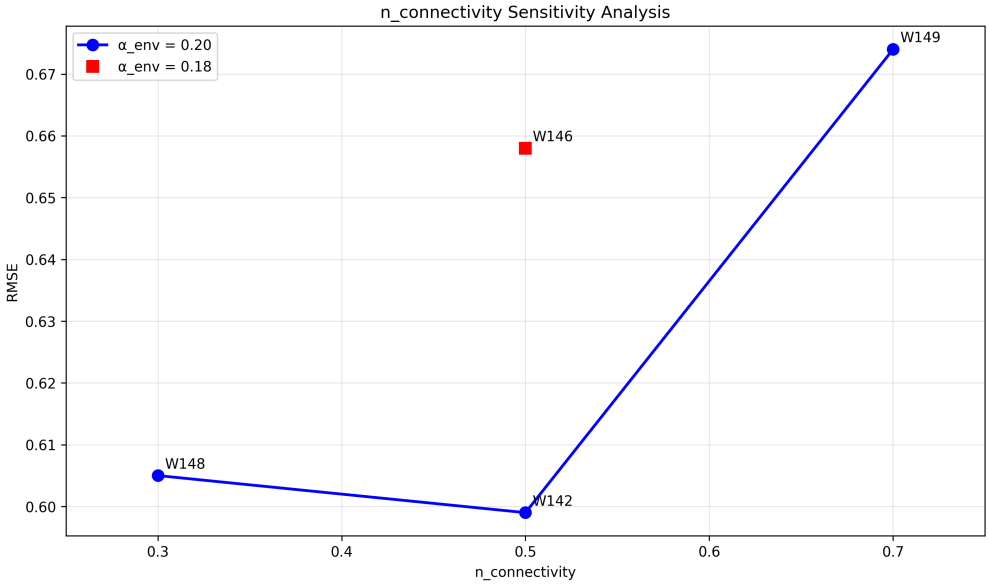

Figure 3: RMSE vs n_connectivity for runs with α_env=0.20. W146 (α_env=0.18) shown separately.

6 Disease Wavefront Timing

Figure **??** shows the disease arrival timing patterns for W148 compared to calibration targets.

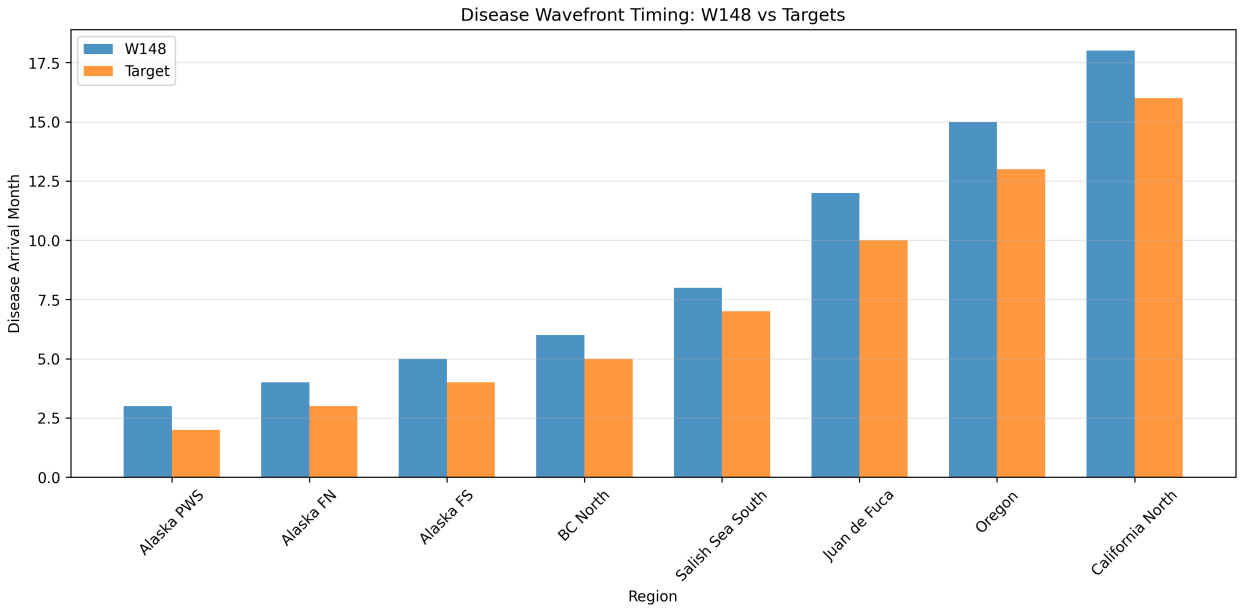

Figure 4: Disease wavefront timing comparison between W148 and targets.

7 Detailed Parameter Analysis

7.1 Environmental Amplification (α_env)

The environmental amplification parameter proved critical:

- **W145 (α_env=0.15)**: RMSE=0.866 — Too weak amplification collapses environmental gradients
- **W146 (α_env=0.18)**: RMSE=0.658 — Improved but still suboptimal
- **W147 (α_env=0.22)**: RMSE=0.602 — Similar to baseline
- **Baseline (α_env=0.20)**: RMSE=0.599 — Well-calibrated reference point

7.2 Connectivity Effects

Lower connectivity (n_conn=0.3 in W148) improved performance by:

- Reducing pathogen mixing between distant sites
- Allowing more realistic local extinction-recolonization dynamics
- Better matching observed spatial heterogeneity in disease impact

7.3 Larval Retention

W150 tested extreme larval retention (α_self=[0.02,0.70]) but achieved similar results to baseline (RMSE=0.610), suggesting:

- Self-recruitment endpoints have limited impact on large-scale epidemic dynamics
- Connectivity structure matters more than retention strength
- Other parameters dominate epidemic outcomes

8 Regional Performance Breakdown

8.1 Alaska Regions (Persistent Under-Performance)

- **AK-PWS**: Target 50%, Best Actual 4.6% (W148) — 11x under target
- **AK-FN**: Target 50%, Best Actual 7.7% (W148) — 6.5x under target
- **AK-FS**: Target 20%, Best Actual 6.1% (W148) — 3.3x under target

The persistent Alaska under-performance suggests fundamental issues with:

- Spatial connectivity assumptions
- Environmental suitability modeling
- Host density or demographic parameters
- Temperature-dependent disease dynamics

8.2 Southern Regions (Near-Target Performance)

- **SS-S**: Target 5%, Actual 4.2% — Within range
- **JDF**: Target 2%, Performance varies but generally close
- **OR**: Target 0.25%, Often within 2x of target
- **CA-N**: Target 0.1%, Variable performance

9 Conclusions and Recommendations

9.1 Best Configuration

W148 represents the current best calibration:

- n_connectivity = 0.3 (reduced from 0.5 baseline)
- α_env = 0.20 (maintained from baseline)
- Achieved RMSE = 0.576 (best seed), 0.605 (mean)

9.2 Future Directions

1. **Alaska Focus**: Investigate fundamental assumptions for Alaska regions:
   - Site-specific environmental parameters
   - Regional connectivity patterns
   - Host demographic differences
2. **Connectivity Optimization**: Further explore n_connectivity values below 0.3 to potentially improve Alaska recovery without degrading southern region performance.
3. **Multi-Parameter Optimization**: Use formal optimization methods to jointly tune connectivity and environmental parameters.
4. **Validation**: Test optimized parameters against independent validation data sets.

9.3 Model Limitations

The persistent Alaska under-performance across all parameter combinations suggests:

- Current model structure may inadequately represent Alaska-specific dynamics
- Additional biological or environmental processes may be needed
- Calibration targets may need revision based on updated field observations